TEXORA · LECTURE SHEET

Dye Synthesis Technology, Plasma Dyeing, Enzymatic Treatment and Placement Printing

Four special technologies in one sheet — how an azo dye is built molecule by molecule, how plasma rewrites a fibre surface, how enzymes replace harsh chemistry, and how a print is engineered to land on exactly one panel of a garment.

WP-4203 · SPECIAL WET PROCESSING · TOPIC 3 · SPECIALIZATION — WET PROCESSING

1. Block A — Dye synthesis technology

1.1 Colour by design: chromophore and auxochrome

A dye molecule is engineered in two parts. The **chromophore** is the unsaturated group whose π-electron system absorbs visible light — azo (–N=N–), anthraquinone, phthalocyanine, methine and others. The **auxochromes** are substituents (–OH, $–NH_2$, $–SO_3H$, –Cl, halogen, alkyl) that do not create colour alone but shift its wavelength (bathochromic = to longer wavelength/deeper; hypsochromic = to shorter), change intensity, and — commercially decisive — confer solubility ($–SO_3H$ as its sodium salt) and fibre affinity or reactivity. Synthesis is the craft of assembling chromophore + auxochromes + affinity/reactive groups into a molecule that is coloured, soluble or dispersible, fibre-substantive, fixable and fast.

1.2 Diazotisation and coupling — the azo workhorse

Azo dyes are the largest dye class. Their synthesis has two signature steps:

Step 1 — Diazotisation. A primary aromatic amine ($Ar–NH_2$) reacts with nitrous acid (generated in situ from $NaNO_2$ + HCl) at **0–5 °C** to give a **diazonium salt** ($Ar–N_2^+\ Cl^-$). The low temperature is not a preference: diazonium salts decompose (with nitrogen loss) as temperature rises, and excess nitrous acid must be controlled (checked with starch–iodide paper; excess destroyed with sulphamic acid/urea before coupling, or it attacks the coupler).

$$Ar–NH_2 + NaNO_2 + 2\ HCl \rightarrow Ar–N_2^+\ Cl^- + NaCl + 2\ H_2O \quad (0–5\ °C)$$

Step 2 — Coupling. The diazonium salt (electrophile) attacks an electron-rich **coupling component** — a phenol, naphthol, aromatic amine or active-methylene compound — usually at the para position to an

–OH/–NH_2 group. Phenolic couplers are coupled in **alkaline** solution (the phenoxide is far more reactive); amine couplers in mildly acidic to neutral conditions. The product carries the new –N=N– chromophore linking the two aromatic systems.

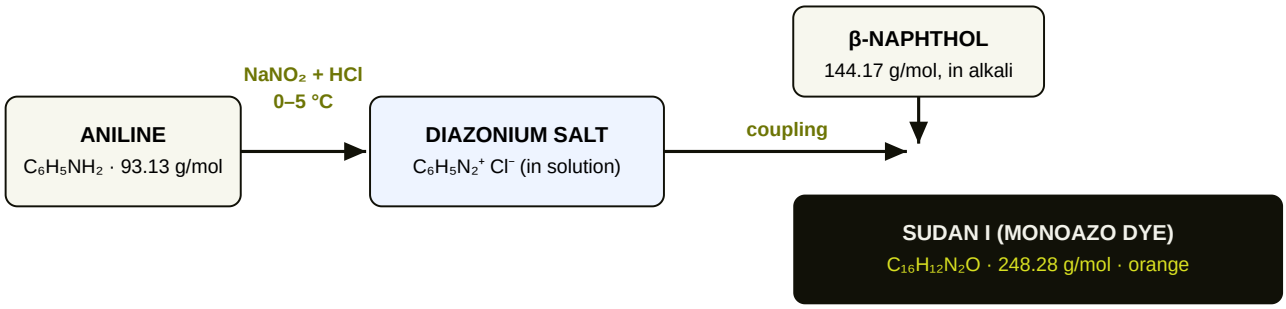


Figure 1 — Diazotisation and coupling scheme: aniline → benzenediazonium chloride → Sudan I by coupling with alkaline β-naphthol. Sudan I is the classic teaching example of a monoazo dye (a solvent dye, not a textile dye for wear — used here for stoichiometry).

WORKED EXAMPLE 1 — AZO SYNTHESIS STOICHIOMETRY (0.10 MOL SCALE)

Aniline, M = 93.13 g/mol: 0.10 × 93.13 = **9.31 g**.

$NaNO_2$, M = 69.00 g/mol: 0.10 × 69.00 = **6.90 g** (1 mol per mol amine; a small controlled excess is used in practice, excess later destroyed).

β-Naphthol, M = 144.17 g/mol: 0.10 × 144.17 = **14.42 g**, dissolved in dilute NaOH for coupling.

Product Sudan I, $C_{16}H_{12}N_2O$, M = 248.28 g/mol: theoretical yield = 0.10 × 248.28 = **24.83 g**.

Actual isolated yield 19.8 g → % yield = 19.8 / 24.83 × 100 = **79.7%**.

Losses arise in diazonium decomposition, incomplete coupling, and filtration/washing — each is a process-control point (temperature 0–5 °C, pH of coupling bath, addition rate).

1.3 Synthesis routes overview — reactive and disperse dyes (concept level)

- **Disperse dyes:** mostly monoazo and anthraquinone structures built by the same diazotisation–coupling chemistry, but finished *without* solubilising sulphonic groups — the product is a hydrophobic solid, milled with dispersing agent to a fine dispersion. Substituents (–CN, –NO_2, halogen) tune shade and sublimation fastness on polyester.
- **Reactive dyes:** a chromophore (azo, anthraquinone or phthalocyanine) is bridged to a **reactive system** — e.g. a chlorotriazine ring or a vinyl-sulphone precursor — and carries sulphonate groups for water solubility. Synthesis adds steps: build/buy the chromophore with an amino handle, condense it with cyanuric chloride under staged temperature/pH control (first chlorine substituted cold, second warmer), then attach the bridging amine/solubiliser. The product must survive storage yet react with cellulose –OH in the dyebath — a stability/reactivity balance set during synthesis.

- **Common plant operations:** diazotisation, coupling, condensation, sulphonation, nitration/reduction to make amine intermediates, salting-out, filtration, standardisation (strength adjustment with diluent), spray drying. Quality is specified as strength (%) against a standard, shade (ΔE), moisture and insolubles — not merely as chemical purity.

2. Block B — Plasma treatment and plasma-assisted dyeing

Plasma is a partially ionised gas — electrons, ions, radicals, excited neutrals and UV — generated by an electric discharge in a gas (air, O_2, N_2, Ar, He). It is sometimes called the fourth state of matter. Crucially for textiles, the *bulk gas stays near room temperature* in "cold" plasma while the reactive species carry enough energy to modify a fibre surface to a depth of only nanometres — the bulk strength and handle of the fabric are untouched.

PLASMA TYPE	CONDITIONS	TEXTILE CHARACTER
Corona discharge	Atmospheric pressure, pointed/roller electrodes, air	Simplest, in-line capable; treatment can be non-uniform; long used for film, adapted for fabrics/nonwovens
Dielectric barrier discharge (DBD)	Atmospheric pressure, dielectric-covered electrodes	More uniform glow; the main atmospheric route for continuous fabric treatment
Low-pressure (glow) plasma	Vacuum chamber, controlled gas (O_2, NH_3, Ar…)	Most uniform and controllable chemistry; batch process, chamber limits fabric size — research and high-value goods

What plasma does to a surface: (i) **cleaning** — oxidises and volatilises surface oils and size residues; (ii) **etching / nano-roughening** — sputtering and radical attack pit the surface, raising area and mechanical keying; (iii) **activation / functionalisation** — grafts polar groups (–OH, –COOH, $–NH_2$ depending on gas) onto the surface; (iv) with monomer vapours, **plasma polymerisation** deposits ultra-thin coatings (hydrophobic, barrier, antistatic). The measurable headline is **wettability change**: water contact angle falls sharply on treated polyester, polypropylene or wool, wicking time drops, and aqueous processes — desizing, scouring, dyeing, printing, coating adhesion — all start from a receptive surface instead of a repellent one.

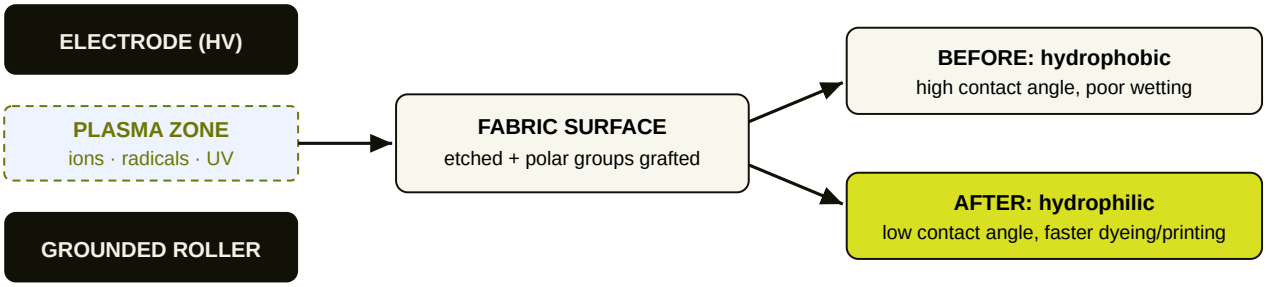


Figure 2 — Plasma surface treatment (DBD/corona geometry). Reactive species clean, etch and functionalise the surface; wettability and adhesion rise without chemicals or water.

"Plasma dyeing" is used in two senses, both built on the above: (1) plasma *pre-treatment* that lets a conventional dye work better — deeper, faster or at lower temperature, better wool shrink-resist/dyeability, dyeing of otherwise undyeable surfaces in research; and (2) direct *plasma coloration* experiments — depositing coloured plasma-polymer films or grafting dye-like species. Sense (1) is the industrially meaningful one today. **Limits:** treatment ages (hydrophobic recovery over days–weeks on some polymers — dye soon after treatment), uniformity over width and through thick assemblies must be proven, and atmospheric systems add electrode/ozone management (ozone is a plasma by-product to be extracted).

3. Block C — Enzymatic treatments

Enzymes are protein catalysts: each acts on a specific substrate, under mild conditions (typically 40–60 °C, near-neutral to mildly acidic/alkaline pH), and is then denatured ("killed") by heat or pH shift to stop the reaction exactly on target. In wet processing they replace hot alkali, oxidants and mechanical harshness with selective biochemistry — lower energy, less fibre damage when controlled, and biodegradable effluent.

ENZYME	SUBSTRATE / JOB	PROCESS NOTES
Amylase	Hydrolyses starch size → soluble dextrins/sugars (desizing)	The classic enzyme process; does not attack cellulose; verify size type first (useless on PVA size)
Cellulase	Hydrolyses surface cellulose — bio-polishing, denim abrasion	Dose/time control weight and strength loss; acid vs neutral cellulases differ in aggressiveness and back-staining
Catalase	Decomposes residual H_2O_2 → water + oxygen after bleaching	Replaces repeated rinses and chemical reducers before dyeing; saves water and time

Laccase	Oxidises phenolic/indigo chromophores (with mediator)	Denim bleaching/washing effects, effluent decolourisation research
Pectinase	Removes pectin — bio-scouring of cotton	Gentler scouring, often combined with other enzymes; absorbency target must be verified
Protease	Protein substrates — wool/silk modification, stain removal	Powerful but risky on protein fibres themselves; tight control needed

WORKED EXAMPLE 2 — CELLULASE BIO-POLISHING WEIGHT LOSS

Fabric lot: 200 g cotton knit.
Controlled bio-polishing target: weight loss 4% (surface fuzz and protruding fibre ends hydrolysed and removed in washing).
Mass removed = 200 × 0.04 = **8.0 g**; finished lot ≈ **192 g** before any moisture-regain normalisation.
Control meaning: at ~3–5% loss the fabric gains a clean, low-pilling, softer surface. Push well beyond this and the enzyme stops eating only fuzz — it thins yarns, and **tensile/tear strength falls disproportionately**. Weight loss is therefore the in-process control: weigh conditioned samples, stop the enzyme (temperature/pH kill) at target, never by clock alone.

Advantages: specificity (amylase touches starch, not cotton), mild conditions (energy saving), less chemical load, improved handle and appearance, biodegradable products. **Limitations:** enzymes are proteins — sensitive to temperature, pH, metal ions and surfactant choice, so process windows are narrow; reaction is progressive, so over-treatment damage (cellulase strength loss) is the standing risk; enzyme cost per kg is higher than commodity chemicals, justified by savings elsewhere; activity varies by storage — assay and date discipline matter; and an enzyme exists only where a substrate matches — no enzyme desizes a synthetic size it cannot hydrolyse.

4. Block D — Placement (engineered) printing

Placement printing — also called **engineered printing** or panel printing — prints a design that is *positioned deliberately on a specific part of a garment*: a chest motif centred on the front panel, artwork that flows exactly across a pocket, a border engineered to sit at a hem, a yoke pattern that aligns at seams. It is the opposite of an all-over repeat: the garment's pattern pieces are the canvas, and print position is part of the product specification, with tolerances (often a few millimetres) written on the order.

4.1 Workflow

1. **Artwork engineering:** the design is laid onto the digital pattern (marker/panel shapes) at graded sizes — a chest print is re-proportioned for each garment size, not merely scaled blindly.
2. **Substrate stage:** printing is done either on *cut panels before sewing* (best registration and flat printing; the dominant route for quality work) or on the *finished garment* (chest/back prints on T-shirts; limited printable area, garment must be flattened on a pallet).
3. **Registration:** panels are aligned to marks, jigs, laser guides or camera systems so the motif lands at the specified distance from neckline, centre front and side seams. Registration = repeatable positioning in X, Y and rotation.
4. **Print, fix, inspect:** fixation (cure for plastisol/pigment, steam/wash for reactive garment printing) is followed by position inspection against the spec — a beautiful print 15 mm off-centre is a reject.

FEATURE	SCREEN PLACEMENT PRINTING	DIGITAL (DTG / INKJET) PLACEMENT
Set-up	One screen per colour; pallet/jig alignment	File + garment pallet; camera alignment on advanced machines
Economics	Wins at long runs of one design	Wins at short runs, many sizes/designs, photographic artwork
Colour / detail	Spot colours, special effects (puff, foil, discharge) strong	Unlimited colours, fine gradients; white underbase needed on dark garments
Registration risk	Panel placement by operator — training-critical	Garment loading still manual on most DTG — same discipline, faster correction

Applications: fashion T-shirts and hoodies (chest/back motifs), sportswear (logos, numbers, sponsor marks at regulated positions), engineered border and panel dresses, children's wear motifs, workwear branding, and all-over-print garments assembled from printed panels where the design must match across seams ("engineered AOP"). Design rules: keep critical artwork away from seams, darts and folds; mirror left/right panels correctly; and confirm the print survives the garment's wash specification at its actual position (chest prints flex and abrade differently from back prints).

5. Glossary — key terms

- **Chromophore:** light-absorbing group that creates colour (e.g. –N=N–).
- **Auxochrome:** substituent modifying shade, intensity, solubility or affinity.
- **Diazotisation:** conversion of $Ar–NH_2$ to a diazonium salt with $NaNO_2$/HCl at 0–5 °C.
- **Coupling component:** electron-rich partner (e.g. β-naphthol) attacked by a diazonium salt to form an azo dye.

- **DBD:** dielectric barrier discharge — atmospheric plasma through dielectric-covered electrodes.
- **Bio-polishing:** cellulase removal of surface fuzz for a clean, soft, low-pilling finish.
- **Registration:** controlled positional alignment of a print to garment landmarks.
- **Engineered print:** design positioned by specification on a garment panel.

6. Key points

- Azo synthesis = diazotise cold (0–5 °C), couple at controlled pH; 0.10 mol worked case gives 24.83 g theoretical, 79.7% at 19.8 g actual.
- Plasma changes only the surface — cleaning, etching, polar grafting — so wettability and dyeability rise with no bulk damage; treatment ages, so process promptly.
- Enzymes are specific and mild but progressive: control by measured effect (e.g. 4% weight loss = 8.0 g on 200 g), then kill the enzyme.
- Placement printing is a positioning specification as much as a printing method; panels before sewing register best.

7. Summary

This sheet joined four special technologies. Dye synthesis builds colour deliberately — chromophore plus auxochromes, azo chemistry by cold diazotisation and pH-controlled coupling, with reactive and disperse dyes as structured extensions; the Sudan I stoichiometry shows how molar bookkeeping becomes yield control. Plasma treatment rewrites fibre surfaces physically and chemically at nanometre depth, raising wettability for dyeing and printing. Enzymatic treatments replace brute-force chemistry with selective catalysis — amylase, cellulase, catalase, laccase — provided their progressive action is measured and stopped on target. Placement printing finally engineers where colour lands on a garment, with registration discipline separating a premium product from a reject.

8. Practice

1. Repeat the azo stoichiometry at 0.25 mol scale: masses of aniline, $NaNO_2$ and β-naphthol, and the theoretical yield of Sudan I.
2. Why must diazotisation be held at 0–5 °C, and why is β-naphthol dissolved in alkali for coupling?
3. A 500 g fabric lot is bio-polished to a controlled 3.5% weight loss. How many grams are removed, and what risk rises if loss reaches 9%?
4. State the three plasma surface effects and name the plasma type best suited to continuous atmospheric fabric treatment.

5. Give two reasons for printing panels before sewing rather than the finished garment, in placement work.

Answer key

1. Aniline: 0.25 × 93.13 = 23.28 g. $NaNO_2$: 0.25 × 69.00 = 17.25 g. β-Naphthol: 0.25 × 144.17 = 36.04 g. Theoretical Sudan I: 0.25 × 248.28 = 62.07 g.
2. Diazonium salts decompose with nitrogen loss as temperature rises, destroying yield and creating side products; cold, acidic conditions stabilise them for the coupling step. In alkali, β-naphthol forms the phenoxide ion, a much stronger electron donor, so electrophilic coupling by the diazonium salt is fast and gives the expected para/ortho substitution.
3. Removed = 500 × 0.035 = 17.5 g. At 9% the cellulase has consumed load-bearing yarn cellulose, not just fuzz — tensile and tear strength fall sharply and the fabric may fail specification.
4. Cleaning (removal of oils/residues), etching/nano-roughening, and activation/functionalisation (grafting polar groups); DBD is the main continuous atmospheric route (corona is also atmospheric but less uniform).
5. Panels lie perfectly flat, giving sharper prints and precise registration to pattern landmarks; and print position relative to seams/neckline can be set from the cut geometry before assembly, whereas a sewn garment distorts on a pallet and limits printable area. (Also acceptable: fixation and inspection are easier before sewing.)
